

**Product Laws of Exponents**Question Bank - Additional 3.01 **ANSWER KEY**

Name

Date

Period

*Show your work. Circle your final answer.***For questions 1 and 2, write the product as a repeated multiplication expression, then as a single power.**

1.  $4^2 \cdot 4^5$

$$(4 \cdot 4)(4 \cdot 4 \cdot 4 \cdot 4 \cdot 4) = 4^7$$

2.  $7^2 \cdot 7^1 \cdot 7^3$

$$7^6$$

$$2 + 1 + 3 = 6.$$

3. Evaluate each power, then multiply.  $5^3 \cdot 5^2$

$$125 \cdot 25 = 3,125$$

*Also  $5^5$ .*

4. Apply the product of powers law, then evaluate the power.  $(-6)^2 \cdot (-6)^7$

$$(-6)^9 = -10,077,696$$

*Nine negative factors: negative.*

5. Leeland says he can evaluate  $7^3 \cdot 3^7$  using the product of powers law. Is he correct? Explain.

**No — the bases are different (7 and 3), so the law does not apply. The value is  $343 \cdot 2,187 = 750,141$ , found by evaluating each power.**

**For questions 6 through 8, use the power of a product law.**

6. Write  $(7 \cdot 8)^2$  as a repeated multiplication expression, group the repeated factors, then write it in the form  $a^m \cdot b^n$ .

$$(7 \cdot 8)(7 \cdot 8) = (7 \cdot 7)(8 \cdot 8) = 7^2 \cdot 8^2$$

*Value 3,136 either way.*

7. Evaluate inside the parentheses first, then apply the exponent.  $(-3 \cdot 6)^4$

$$(-18)^4 = 104,976$$

*Math Nation's key prints 104,796 for this item; it is 104,976.*

8. Apply the power of a product law, then evaluate.  $(2 \cdot (-9))^3$

$$2^3 \cdot (-9)^3 = 8 \cdot (-729) = -5,832$$

For questions 9 and 10, use the power of a power law or the identity law.

9. Write  $[(5)^4]^4$  as a single power, then give its value.

$$5^{16} = 152,587,890,625$$

10. Rewrite  $[(2)^5]^2$  using a law of exponents, name the law, then evaluate.

$$2^{10} = 1,024 \quad \text{— power of a power}$$

*Math Nation's key stops at  $2^{10}$ ; the question asks for the value.*

**Working backwards.**

11. Find the value of  $n$  that makes each statement true.

a.  $3^n \cdot 3^4 = 3^9$

$$n = 5$$

b.  $(2^n)^3 = 2^{12}$

$$n = 4$$

c.  $\left(\frac{1}{2}\right)^n = \frac{1}{32}$

$$n = 5$$

12. Which expression is equivalent to  $(3^4)^2$  ?

☒ A.  $3^8$

☐ B.  $3^6$

☐ C.  $2 \cdot 3^4$

☐ D.  $3^{16}$

**A**

*Multiply the exponents.*

13. Select ALL of the expressions that are equivalent to  $10^6$  .

☒ **A.**  $10^3 \cdot 10^3$

☒ **B.**  $(10^2)^3$

☒ **C.**  $10^2 \cdot 10^4$

☒ **D.**  $100^3$

☐ **E.**  $10^2 \cdot 10^3$

☐ **F.**  $(10^3)^3$

***A, B, C and D***

*D:  $100^3 = 1,000,000 = 10^6$ .*